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Experiences

Research and Development Engineer, [VN Instruments](#). August 2024 – Present.

- Leading all technical aspects of the Martian saltation sensor development program in collaboration with NASA Ames. Completed 5 of 9 major milestones for the 7-year program in the last 14 months.
- Delivered testbed capable of shooting individual sub-millimeter microspheres at prototype sensors in a vacuum chamber, while measuring microsphere mass and speed using custom optics system.
- Responsible for hardware integration of all commercial applications, prototyping of novel sensor concepts and experimental setups for planetary science applications. Shipped ~2 mini-projects per week.
- Decreased R&D costs by ~16dB\$ and increased iteration speed by 3X.
- Hired and managed 2-3 new interns every 4 months.

Propulsion Subsystem Lead, [Waterloo Rocketry](#). June 2021 – July 2023.

- Led development of hybrid rocket's oxidizer venting system (MEOP 1,000psi) for 2 competition cycles.
- Responsible for execution of entire development life cycle of a second-generation custom oxidizer vent valve for controlled venting of the launch vehicle's oxidizer tank.
- Machined 15+ custom parts, including aluminum airframe couplers to join critical vehicle sections.
- Participated in numerous engine static fires, wet dress rehearsals, launch rehearsals, and countless other late-night tests to successfully launch the 2023 rocket to 31,476ft on the first launch attempt. Served as launch pad operator during launch.

Radon Assay Research Intern, [SNOLAB](#). September 2022 – December 2022.

- Designed, built, and operated high-vacuum radon assay systems that interfaced with live experimental volumes in the world's deepest cleanroom particle physics laboratory.

Projects

Spray Foam Robot. October 2024 – November 2024.

Weekends project with coworkers. Built a 5m tall, 2-axis vertical gantry with a spray foam hose to spray walls.

- Responsible for developing the entire control system stack – PLC programming, system design, sourcing, integration, testing.
- Welded the torch and welder to manipulate steel into gantry stages.

Spin Stabilization System for Cable-Hoisted Payloads (SpinStop). August 2023 – March 2024.

Final university team project. Focused on stabilizing rescue subjects hoisted by helicopters.

- Responsible for reaction wheel and adaptors design and machining, design space optimization via MATLAB, control system design and testing, prototype development, and project pitching.
- Contributed to mechanical design of 5+ critical parts via SolidWorks, system harnessing, vendors and parts sourcing, and overall systems integration and testing.
- Delivered a functional full-scale prototype with live demos. Won a \$12k pitch competition, Best Project Presentation Award, and nominated for Best Overall Project.

Autonomous Wall-Climbing Vehicle (Wall-E). January 2023 – March 2023.

Team Project. Magnets-tracked vehicle that autonomously climbs over a steel wall and locates a target.

- Responsible for chassis and drive train design (SolidWorks), systems integration. It climbed the wall!

Education

BASc. Mechatronics Engineering, University of Waterloo. 2024.